



BBEE103-203 Question Paper

Basic Electronics (Visvesvaraya Technological University)



Scan to open on Studocu

MAKE-UP EXAM

USN

BBE103 203

First/Second Semester B.E./B.Tech. Degree Examination, Nov./Dec. 2023

Basic Electronics for EEE Stream

Time: 3 hrs.

MAN MARRS . . .

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. Assume any missing data suitably.*

			Module – 1		
Q.1	a.	With the help of neat circuit diagram and waveform, explain the working of full wave rectifier with center tapped transformer.	8	1.2	CO1
	b.	Explain the forward and reverse characteristics of semiconductor diode	6	1.2	CO1
	c.	A Zener regulator has the following data : $V_Z = 16V$, $V_Z = 6V$, $I_{Zmax} = 60mA$, $Z_Z = 7\Omega$, $R = 150\Omega$. Calculate line regulation and ripple rejection ratio.	6	1.3	CO1
OR					
Q.2	a.	Describe the working of a capacitor filter for a half wave rectifier with a neat circuit diagram and necessary waveforms.	8	1.3	CO1
	b.	What is a DC load line? With the help of neat circuit diagram and waveform, explain the procedure of constructing a DC load line for a semiconductor diode.	8	1.3	CO1
	c.	Define : i) Line regulation ii) Load regulation.	4	1.2	CO1
Module – 2					
Q.3	a.	Explain the construction and working of n-channel JFET.	8	1.3	CO2
	b.	Draw and explain output characteristics of CE configuration.	6	1.3	CO2
	c.	For the base bias circuit, $R_C = 12K\Omega$, $R_B = 470K\Omega$, $V_{CC} = 20V$ and $V_{BE} = 0.7V$. Construct the DC load line and indicate the values.	6	1.3	CO2
OR					
Q.4	a.	Explain the construction and working of n-enhancement MOSFET	8	1.3	CO2
	b.	Explain the common base output characteristics.	8	1.2	CO2
	c.	Describe how a transistor can be used as a voltage amplifier.	4	1.2	CO2
Module – 3					
Q.5	a.	Explain the block diagram of a typical op-amp.	8	1.2	C2
	b.	Explain the following terms : i) Input offset current ii) Input bias current iii) CMRR iv) Slew rate v) Input offset voltage vi) Voltage gain.	6	1.2	C2

	c.	Construct an adder circuit using op-amp to obtain an output voltage $V_o = [2V_1 + 3V_2 + 5V_3]$	6	13	CO2
--	----	--	---	----	-----

OR

Q.6	a.	Describe an integrating amplifier using an op-amp in an inverting configuration.	8	12	CO2
	b.	Explain basic differential amplifier using op-amp.	8	12	CO2
	c.	List the characteristics of an ideal op-amp.	4	12	CO2

Module - 4

Q.7	a.	Explain full adder circuit with truth table. Realize the circuit for sum and carry using basic gates.	8	12	CO3
	b.	Explain the Boolean function $F = A + BC$ in a sum of minterms	6	13	CO3
	c.	Convert the following i) $(10AB)_{10} = (?)_{10}$ ii) $(240)_{10} = (?)_2$ iii) $(1234.56)_8 = (?)_{10}$	6	12	CO3

OR

Q.8	a.	Simplify the following Boolean expressions : i) $f(w, x, y, z) = x + xyz + xyz + wx + wx + xy$ ii) $f = (A + B + C)(A + B + C) + (A + B)$	8	13	CO3
	b.	Explain SOP and POS with examples.	6	13	CO3
	c.	Implement half adder using basic gates.	6	12	CO3

Module - 5

Q.9	a.	With suitable diagram, explain working of Linear variable differential transducer.	8	12	CO4
	b.	What is modulation? Describe the need of modulation in communication systems.	6	12	CO4
	c.	Write short note on piezoelectric transducer.	6	12	CO4

OR

Q.10	a.	Explain the working of potentiometric resistive transducer with neat diagram.	8	12	CO5
	b.	Explain the various blocks involved in an electrical communication systems.	6	12	CO5
	c.	A parallel plate capacitive transducer has a plate area $(l \times w) = 40\text{mm}^2$, 40mm and plate spacing $(d) = 0.5\text{mm}$. Calculate the device capacitance and the displacement that causes the capacitance to change by 5pF . Also determine the transducer sensitivity.	6	13	CO5
